SMART HOME MANAGEMENT SYSTEM USING IOT

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Branch : Computer Science

Subject : Creative And Innovative Project

Sub Code : XC 5811

Date : 17 Feb 2026

ABSTRACT

This project develops an IoT-based home management system using an ESP32 microcontroller to control light and fan. Android application is used for wireless control through Wi-Fi, while an ultrasonic sensor automatically controls the light based on human presence. The system improves convenience and energy efficiency and demonstrates a simple and effective smart home solution.

MODULES

1. Sensor Interface Module

- HC-SR04 ultrasonic sensor integration
- Distance calculation & presence detection
- Noise filtering logic

2. Appliance Control Module

- Relay-based ON/OFF control
- Safe AC switching
- Manual override support

3. Wi-Fi Communication Module

- ESP32 Wi-Fi setup
- HTTP communication
- Auto reconnection handling

MODULES

4. Command Processing Module

- Process ON/OFF requests
- Manual + Automatic mode logic
- State synchronization

5. Android Application Module

- Simple ON/OFF UI
- Real-time status update
- Wi-Fi communication with ESP32

TESTING STRATEGY

1. Unit Testing

- Sensor accuracy
- Relay switching
- Wi-Fi connectivity
- Command validation

2. Integration Testing

- App ↔ ESP32 communication
- Sensor + Control logic testing

3. System Testing

- Complete workflow validation
- Network recovery
- Performance efficiency

Test Cases

Test Type	Test Case	Input	Expected Output
Unit Test	Verify Wi-Fi connection	Valid SSID & Password	WiFi connected, IP displayed
Unit Test	Validate ultrasonic distance reading	Object at 10 cm	Distance ≈ 10 cm displayed
Unit Test	Test presence detection logic	Distance < threshold	Light turns ON
Unit Test	Test no presence logic	Distance > threshold	Light turns OFF
Unit Test	Verify relay switching	Send ON command	Relay activates, appliance ON
Unit Test	Test command parsing	HTTP ON/OFF request	Correct appliance toggled
Integration Test	Test app-to-ESP32 communication	Button press ON	Appliance turns ON within 2–5 sec
Integration Test	Test automatic mode with manual override	Presence + Manual OFF	System follows priority logic
System Test	End-to-end workflow	Power ON system	Boot → WiFi → Ready → Control works

CONCLUSION

The second review successfully finalizes the detailed design of the IoT-based Smart Home Management System using ESP32. All core modules including sensor interfacing, relay control, Wi-Fi communication, command processing, and Android application integration have been completed. Manual and automatic control modes are clearly defined, and comprehensive testing strategies are established. The system is now ready for hardware implementation, integration, and real-time validation

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Thank you!